MMWEC PROJECT SUMMARY

Forecasting ISO New England Peak Demand

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460 MW
mean absolute error across 2022 through 2024 validation

12 of 12
monthly peaks captured with 3 dispatches per month

421 MW
2025 daily peak mean absolute error

OBJECTIVE

Can we create a prototype model using public ISO New England load and regional weather forecasts to predict each day's peak demand and identify the days most likely to contain the monthly system peak?

APPROACH

I combined official ISO New England hourly system load with GFS temperature forecasts issued 24 hours before each valid hour. Forecasts were averaged across Boston, Hartford, Providence, Concord, Portland, and Burlington. The model uses nonlinear temperature effects, a weighted measure of forecast temperature across three days, holidays, and calendar seasonality. Historical load features include the daily peak from 1, 2, 3, 7, 14, 21, and 28 days earlier; averages over the trailing 3, 7, 14, and 28 days; and the variability and maximum peak over the prior 3, 14, and 28 days. All load features are shifted so that no future load information enters a forecast.

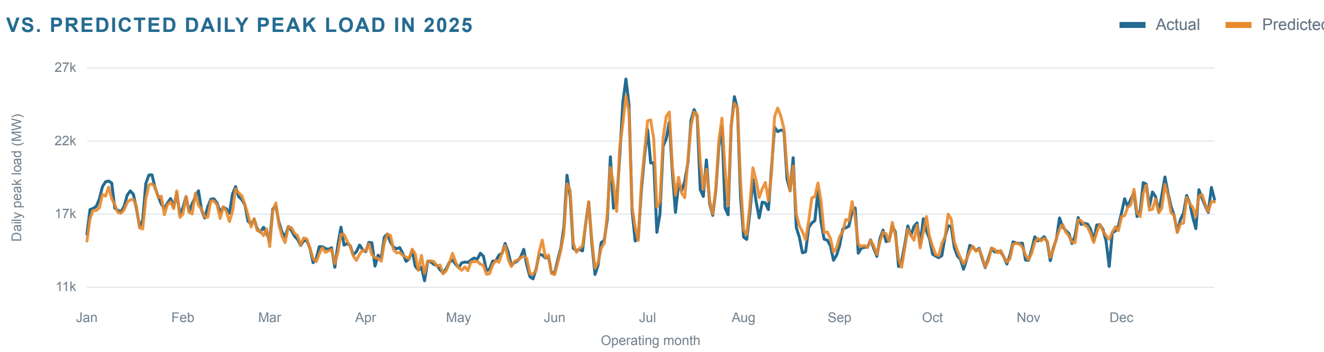
EVALUATION

I compared several models using validation with expanding training windows to determine which performed best across 2022 through 2024. Each validation year was predicted using only data from prior years. After selecting the best model, I evaluated it on 2025 data reserved for testing and not used to fit the model. The selected ensemble combines 75% gradient boosting with 25% Ridge regression. It had the lowest validation error and achieved a 421 MW MAE in the retrospective 2025 evaluation.

FROM FORECAST TO DISPATCH

For each completed month in 2025, I ranked days by predicted peak. The single highest forecast captured 8 of 12 monthly peaks, while two candidate days captured all 12. A third selection added no benefit in this backtest.

ACTUAL VS. PREDICTED DAILY PEAK LOAD IN 2025



Retrospective 2025 Evaluation: The model follows seasonal and daily demand patterns well. The model very rarley over predicts daily peak load.

WHAT THE RESULT SHOWS

In a live setting, the model could estimate tomorrow's peak demand each day using the latest weather forecast and recent load. An operator could compare that estimate with demand already observed during the month and a threshold that reflects the days and dispatch opportunities remaining. If tomorrow appears likely to set the monthly peak and the battery is available, the operator could schedule a discharge once a separate model identifies the likely peak hour.

LIMITATIONS

The model predicts total electricity demand across New England rather than demand for a specific municipal utility. It predicts daily peak magnitude and ranks likely peak days, but it does not predict the hour of the peak. The dispatch test selects each month's three highest forecasts after the month is complete. A real operating strategy would need to decide whether to dispatch each day without knowing the forecasts or conditions for the rest of the month. The test also does not consider operating costs or limititations on dispatch such as charging time, state of charge, degradation, outages, or maximum cycle limits.

FUTURE IMPROVEMENTS

Given more time, I would first add the actual archived values for variables such as humidity, dew point, and temperature in addition to their forecasts. Also, I would build a separate model for the intraday peak hour, and produce prediction intervals rather than a single point forecast. I would then test a sequential dispatch rule using load data from the utility, battery state of charge, degradation costs, the billing rules that determine which regional peaks affect the utility's costs, and any other the information available before each operating decision.

Sources: ISO New England System Loads in EEI Format and Open Meteo Previous Runs API.